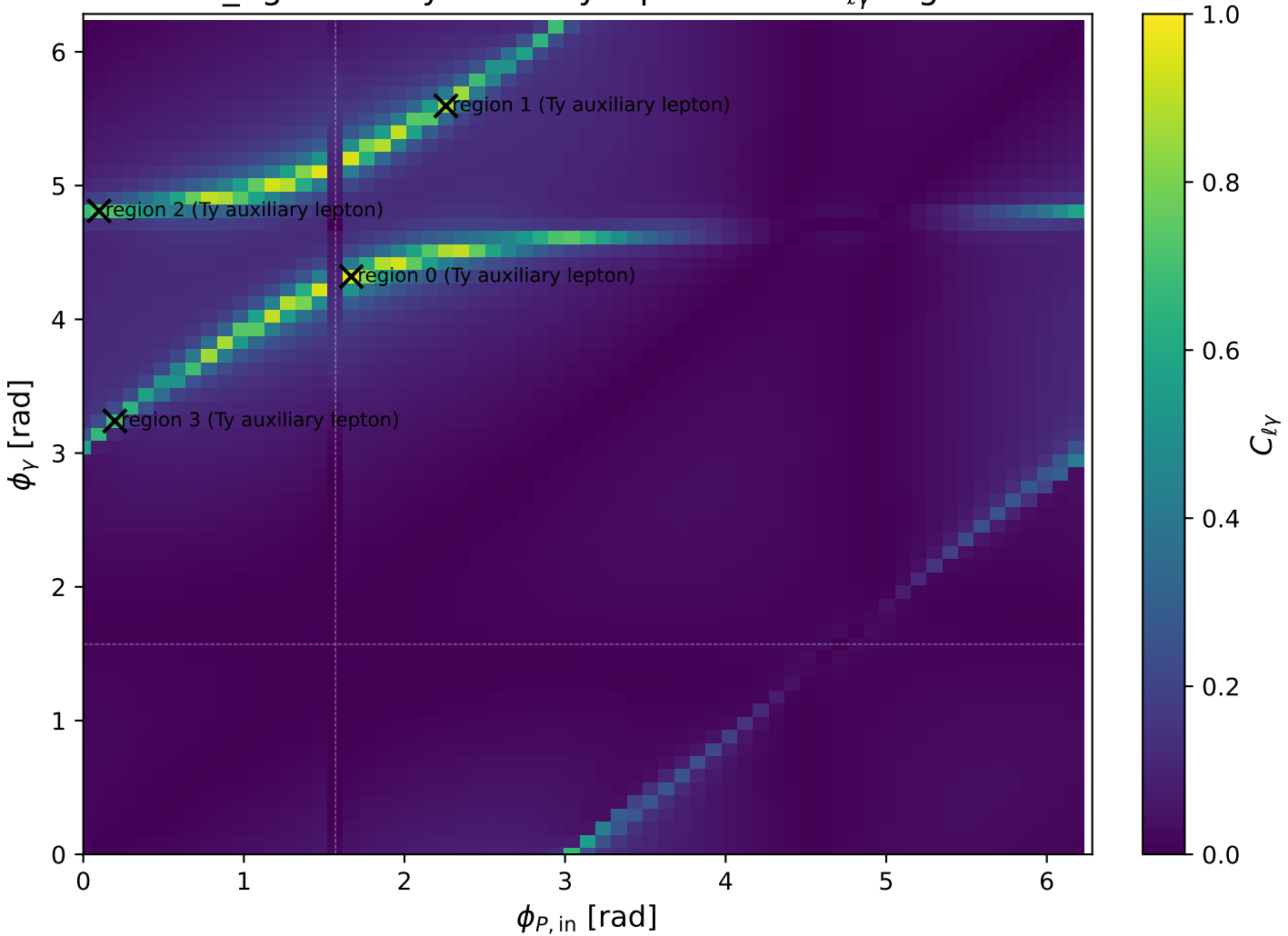
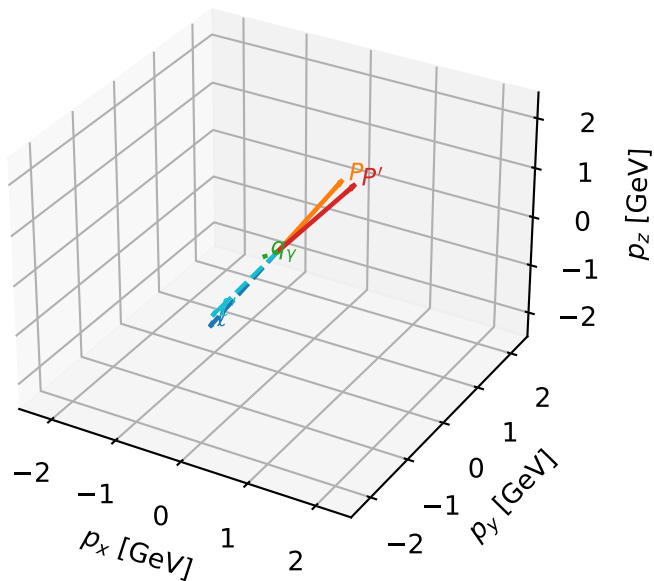


low\_Egamma Ty auxiliary lepton: max  $C_{\ell\gamma}$  regions



# Ty auxiliary lepton characteristic max $C_{\ell\gamma}$ configuration

3D momenta

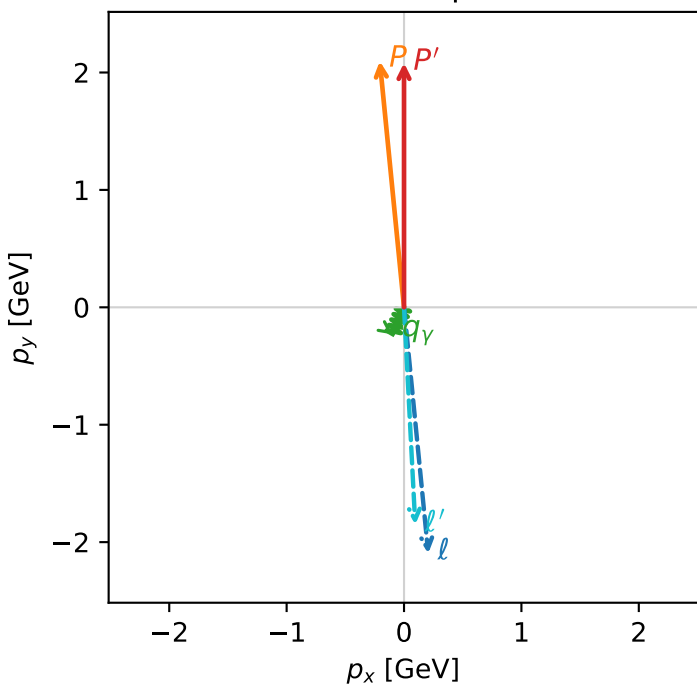


c\_aux\_gamma\_low\_egamma\_ty\_lepton\_region\_0 C\_{l\_aux\gamma}=0.949701  
 region: low\_Egamma\_Ty\_lepton\_region\_0 final-pair delta\_xy=0.444355 rad  
 outgoing-state purity: 0.500746  
 kinematic point: high\_s\_high\_theta\_in\_low\_Egamma  
 incoming state:  $\frac{1}{2}\sum_{h_p}\rho(T_{\gamma,l},h_p)$

$s=21.5158$ ,  $\sqrt{s}=4.63852$ ,  $|\vec{P}|=2.10694$ ,  $|\vec{P}'|=2.0814$   
 $\theta_{in}=1.57079$ ,  $\phi_l=4.81056$ ,  $\phi_P=1.66897$   
 $E_\gamma=0.25$ ,  $\phi_\gamma=4.31969$   
 $Q^2=0.0215983$ ,  $x_B=0.0101659$ ,  $t=-0.0423427$ ,  $W^2=2.98282$ ,  $y=0.108198$   
 $\theta(l',\gamma)=25.4597$  deg

four-momenta  $[E, p_x, p_y, p_z]$  GeV:  
 $l$   $E= 2.3322$   $p_x= 0.2065$   $p_y= -2.0968$   $p_z= -0.0000$   
 $P$   $E= 2.3063$   $p_x= -0.2065$   $p_y= 2.0968$   $p_z= 0.0000$   
 $l'$   $E= 2.1055$   $p_x= 0.0957$   $p_y= -1.8504$   $p_z= 0.0000$   
 $P'$   $E= 2.2830$   $p_x= 0.0000$   $p_y= 2.0814$   $p_z= 0.0000$   
 $q_\gamma$   $E= 0.2500$   $p_x= -0.0957$   $p_y= -0.2310$   $p_z= 0.0000$

Transverse plane



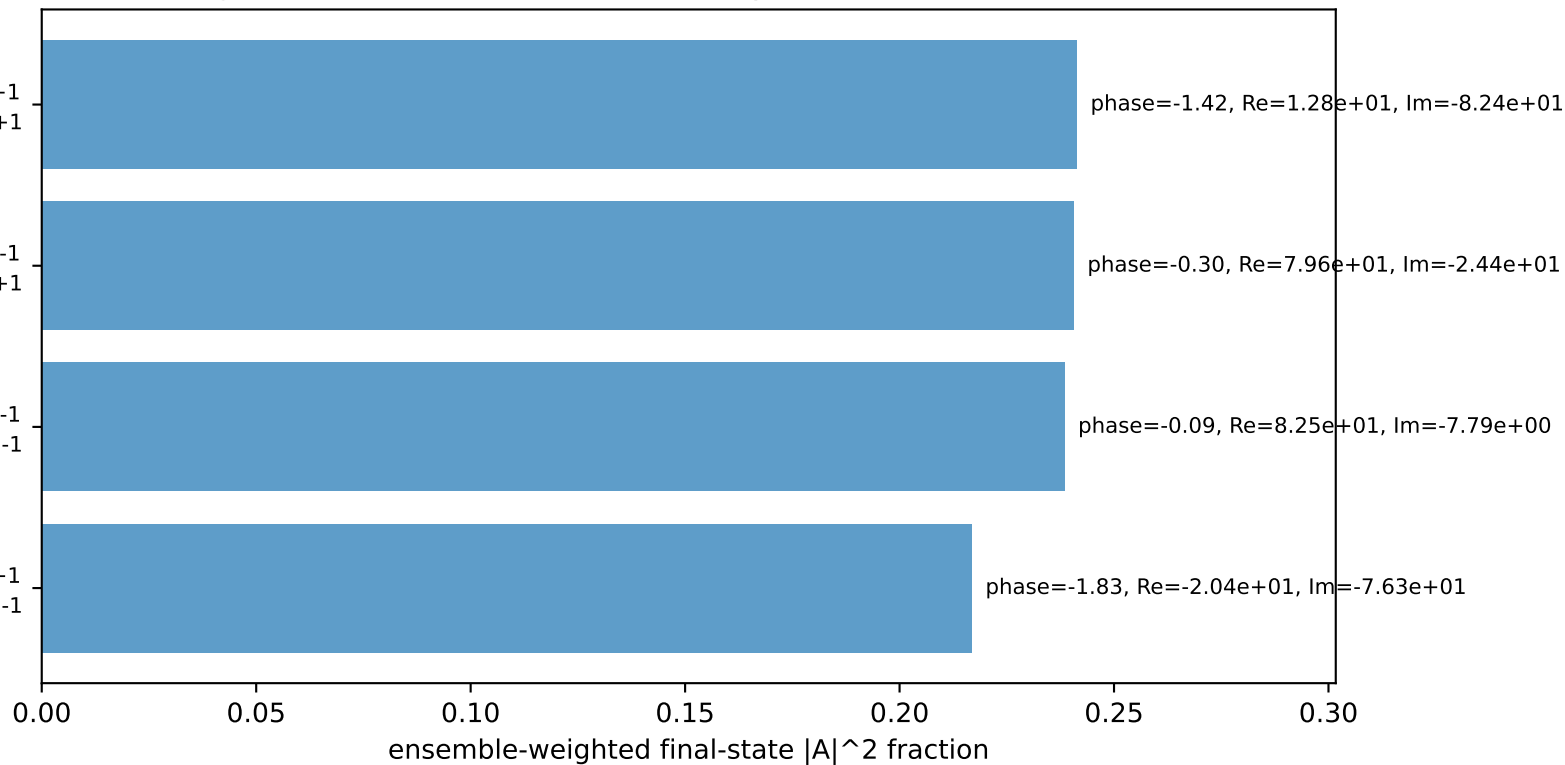
$h_e=-1, h_p=+1, h_\gamma=+1$   
 electron Ty, proton h=+1

$h_e=+1, h_p=+1, h_\gamma=-1$   
 electron Ty, proton h=+1

$h_e=+1, h_p=-1, h_\gamma=-1$   
 electron Ty, proton h=-1

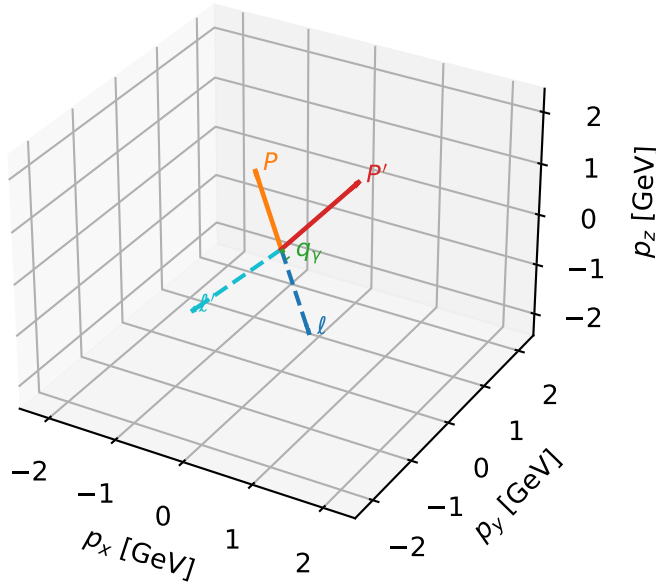
$h_e=-1, h_p=-1, h_\gamma=+1$   
 electron Ty, proton h=-1

Leading initial-ensemble/final-state components (N=4, retained=93.7%)



# Ty auxiliary lepton characteristic max $C_{\ell\gamma}$ configuration

3D momenta



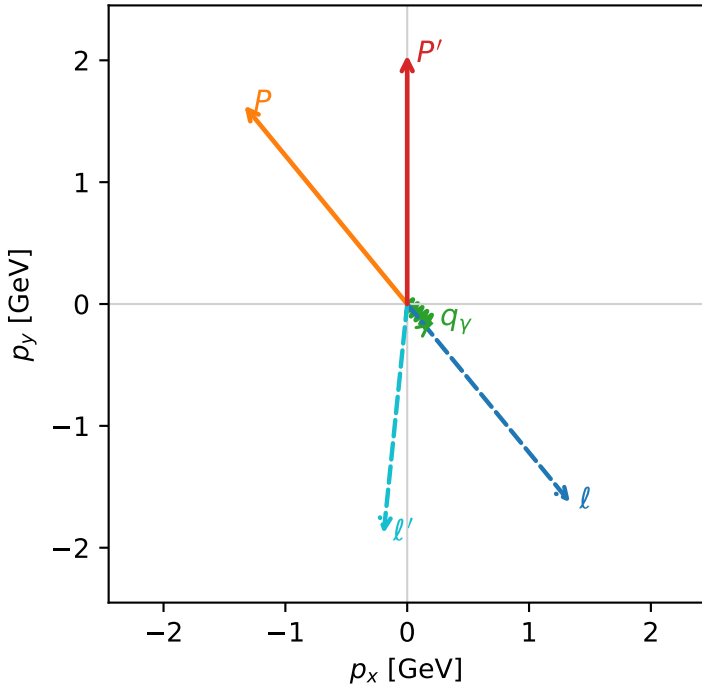
c\_aux\_gamma\_low\_egamma\_ty\_lepton\_region\_1  $C_{\ell\gamma}$ =0.868243  
 region: low\_Egamma\_Ty\_lepton\_region\_1 final-pair delta\_xy=0.985816 rad  
 outgoing-state purity: 0.505556  
 kinematic point: high\_s\_high\_theta\_in\_low\_Egamma  
 incoming state:  $\frac{1}{2}\sum_{h_p}\rho(T_{y,\ell},h_p)$

$s=21.5158$ ,  $\sqrt{s}=4.63852$ ,  $|\vec{P}|=2.10694$ ,  $|\vec{P}'|=2.04213$   
 $\theta_{in}=1.57079$ ,  $\phi_l=5.39961$ ,  $\phi_P=2.25802$   
 $E_\gamma=0.25$ ,  $\phi_\gamma=5.59596$   
 $Q^2=2.36903$ ,  $x_B=0.572172$ ,  $t=-1.95402$ ,  $W^2=2.65122$ ,  $y=0.210858$   
 $\theta(l',\gamma)=56.4831$  deg

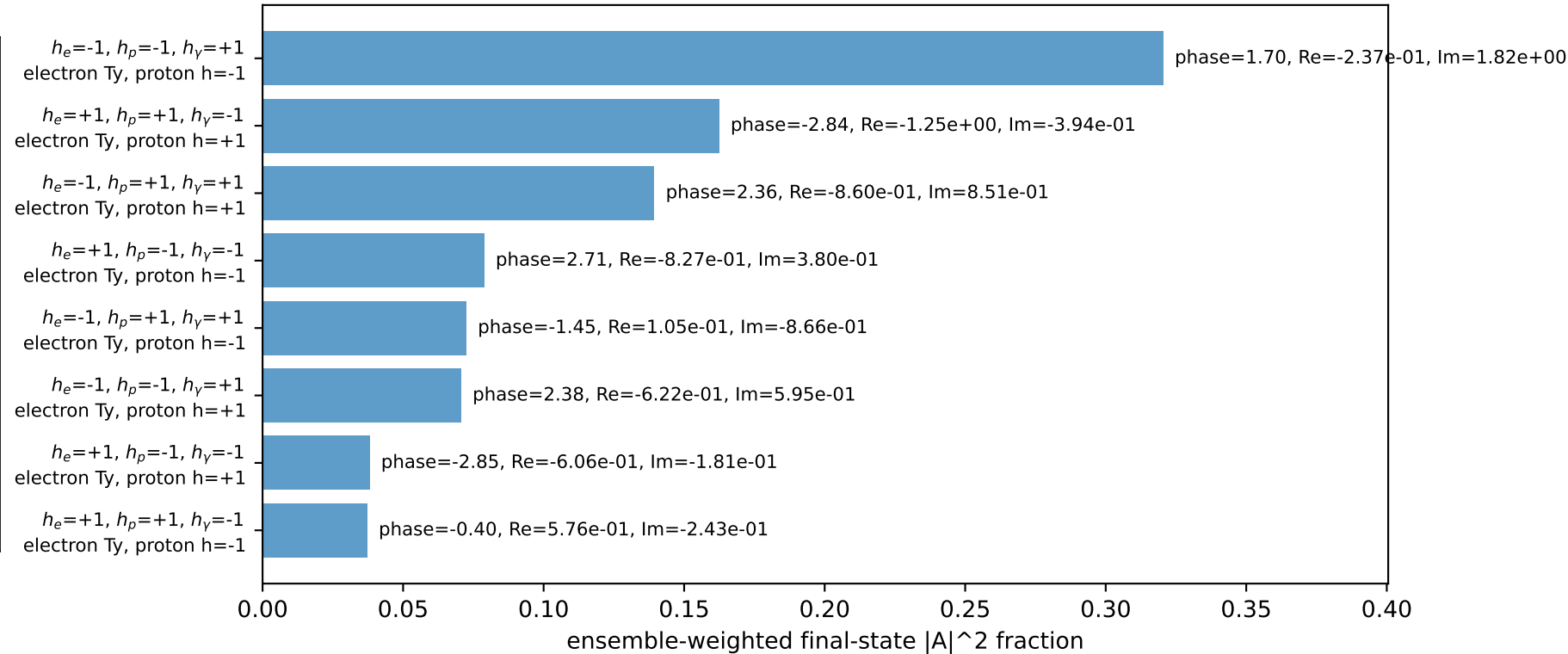
four-momenta  $[E, p_x, p_y, p_z]$  GeV:

| Particle   | E      | $p_x$   | $p_y$   | $p_z$   |
|------------|--------|---------|---------|---------|
| $l$        | 2.3322 | 1.3366  | -1.6287 | -0.0000 |
| $P$        | 2.3063 | -1.3366 | 1.6287  | 0.0000  |
| $l'$       | 2.1413 | -0.1933 | -1.8835 | 0.0000  |
| $P'$       | 2.2472 | 0.0000  | 2.0421  | 0.0000  |
| $q_\gamma$ | 0.2500 | 0.1933  | -0.1586 | 0.0000  |

Transverse plane

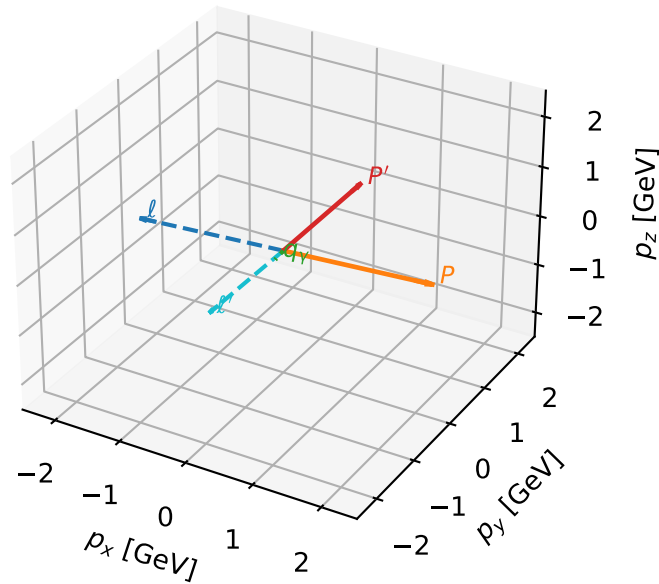


Leading initial-ensemble/final-state components (N=8, retained=91.9%)



# Ty auxiliary lepton characteristic max $C_{\ell\gamma}$ configuration

3D momenta

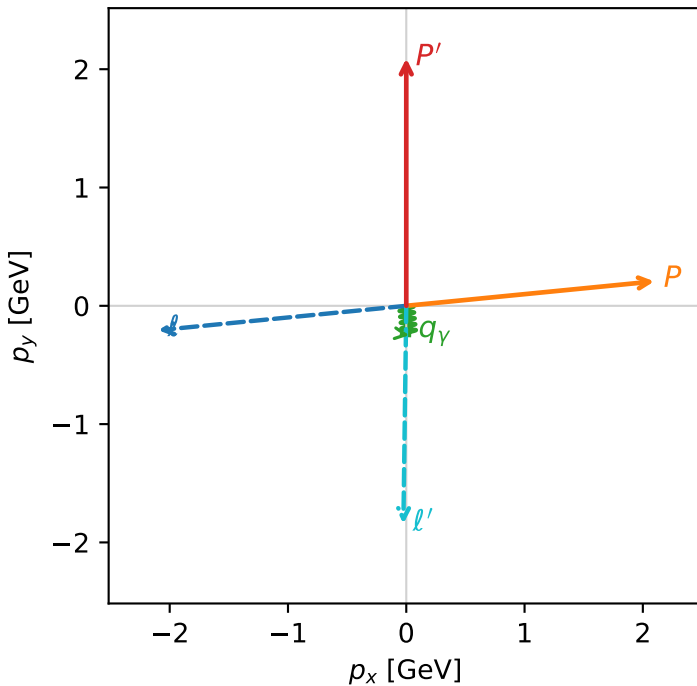


c\_aux\_gamma\_low\_egamma\_ty\_lepton\_region\_2  $C_{\ell\gamma}$ =0.734918  
 region: low\_Egamma\_Ty\_lepton\_region\_2 final-pair delta\_xy=0.111474 rad  
 outgoing-state purity: 0.543446  
 kinematic point: high\_s\_high\_theta\_in\_low\_Egamma  
 incoming state:  $\frac{1}{2}\sum_{h_p}\rho(T_{y,\ell},h_p)$

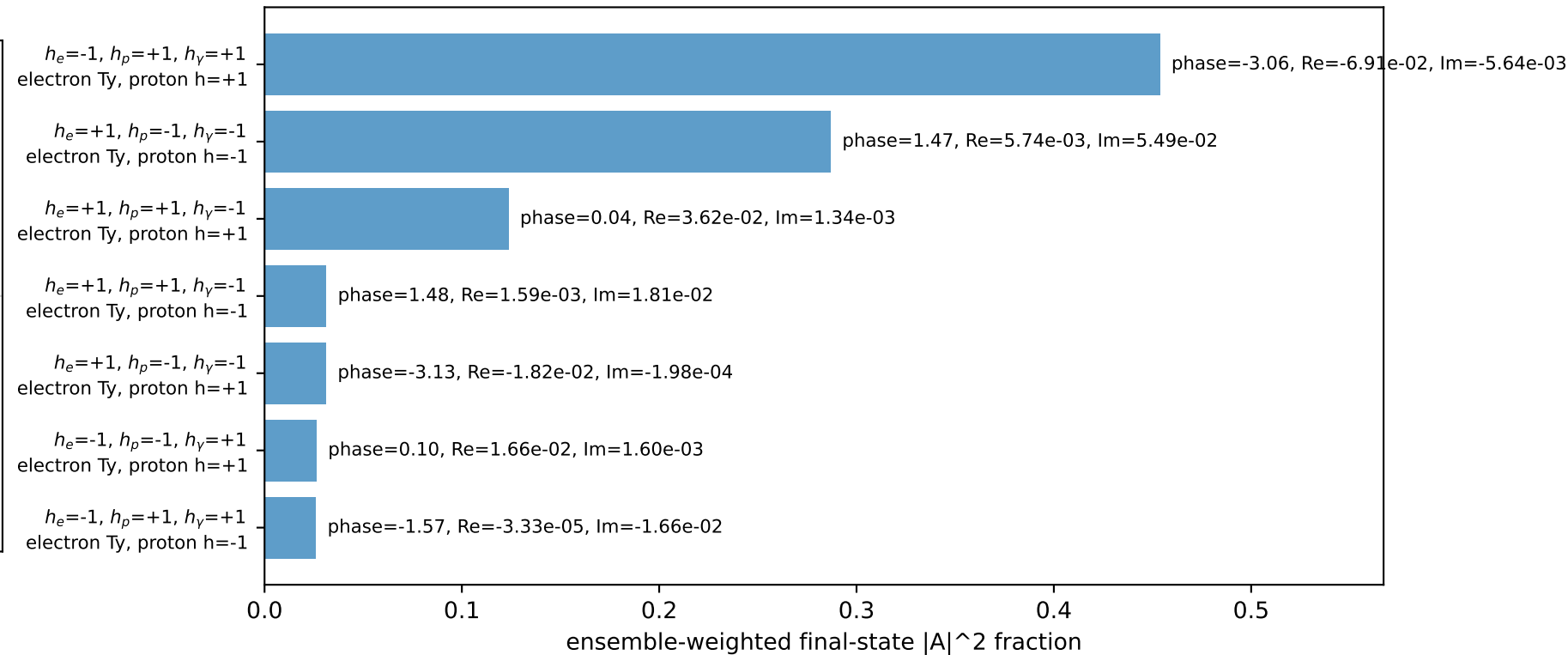
$s=21.5158$ ,  $\sqrt{s}=4.63852$ ,  $|\vec{P}|=2.10694$ ,  $|\vec{P}'|=2.09128$   
 $\theta_{in}=1.57079$ ,  $\phi_l=3.23977$ ,  $\phi_P=0.0981748$   
 $E_\gamma=0.25$ ,  $\phi_\gamma=4.81056$   
 $Q^2=6.91524$ ,  $x_B=0.759762$ ,  $t=-7.94869$ ,  $W^2=3.06645$ ,  $y=0.463528$   
 $\theta(\ell',\gamma)=6.38697$  deg

four-momenta  $[E, p_x, p_y, p_z]$  GeV:  
 $\ell$   $E= 2.3322$   $p_x= -2.0968$   $p_y= -0.2065$   $p_z= -0.0000$   
 $P$   $E= 2.3063$   $p_x= 2.0968$   $p_y= 0.2065$   $p_z= 0.0000$   
 $\ell'$   $E= 2.0965$   $p_x= -0.0245$   $p_y= -1.8425$   $p_z= 0.0000$   
 $P'$   $E= 2.2920$   $p_x= 0.0000$   $p_y= 2.0913$   $p_z= 0.0000$   
 $q_\gamma$   $E= 0.2500$   $p_x= 0.0245$   $p_y= -0.2488$   $p_z= 0.0000$

Transverse plane

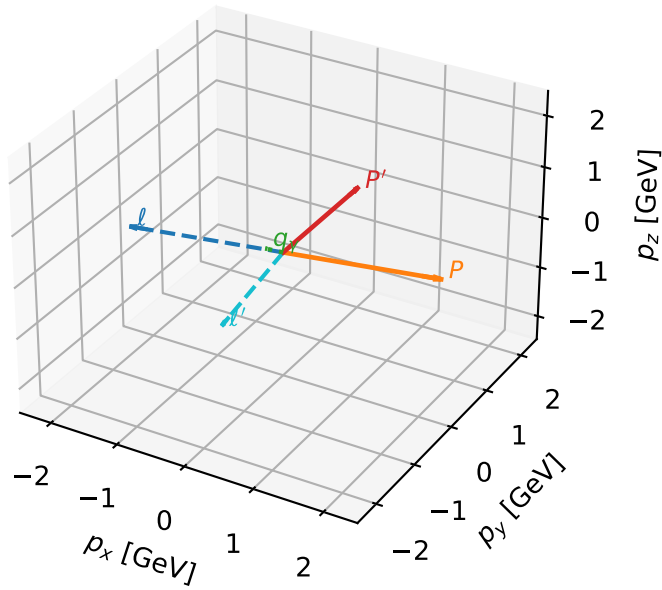


Leading initial-ensemble/final-state components (N=7, retained=97.9%)



# Ty auxiliary lepton characteristic max $C_{\ell\gamma}$ configuration

3D momenta

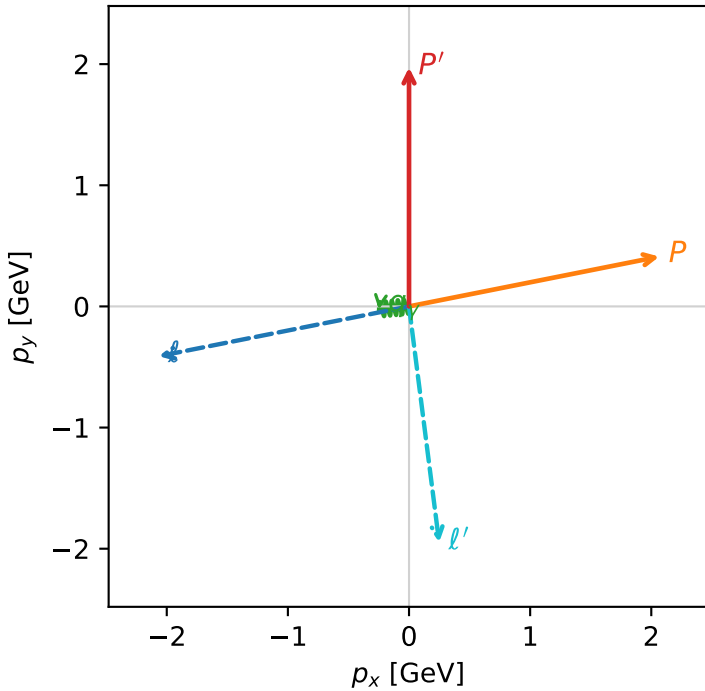


c\_aux\_gamma\_low\_egamma\_ty\_lepton\_region\_3  $C_{\ell\text{aux}\gamma}=0.686869$   
 region: low\_Egamma\_Ty\_lepton\_region\_3 final-pair delta\_xy=1.59963 rad  
 outgoing-state purity: 0.529576  
 kinematic point: high\_s\_high\_theta\_in\_low\_Egamma  
 incoming state:  $\frac{1}{2}\sum_{h_p}\rho(T_{y,\ell},h_p)$

$s=21.5158$ ,  $\sqrt{s}=4.63852$ ,  $|\vec{P}|=2.10694$ ,  $|\vec{P}'|=1.97282$   
 $\theta_{\text{in}}=1.57079$ ,  $\phi_{\ell}=3.33794$ ,  $\phi_P=0.19635$   
 $E_{\gamma}=0.25$ ,  $\phi_{\gamma}=3.23977$   
 $Q^2=7.7072$ ,  $x_B=0.866355$ ,  $t=-6.69456$ ,  $W^2=2.06876$ ,  $y=0.453051$   
 $\theta(\ell',\gamma)=91.6522$  deg

four-momenta  $[E, p_x, p_y, p_z]$  GeV:  
 $\ell$   $E=2.3322$   $p_x=-2.0665$   $p_y=-0.4110$   $p_z=-0.0000$   
 $P$   $E=2.3063$   $p_x=2.0665$   $p_y=0.4110$   $p_z=0.0000$   
 $\ell'$   $E=2.2041$   $p_x=0.2488$   $p_y=-1.9483$   $p_z=0.0000$   
 $P'$   $E=2.1845$   $p_x=0.0000$   $p_y=1.9728$   $p_z=0.0000$   
 $q_{\gamma}$   $E=0.2500$   $p_x=-0.2488$   $p_y=-0.0245$   $p_z=0.0000$

Transverse plane



Leading initial-ensemble/final-state components (N=7, retained=93.7%)

